

Nmap Reconnaissance Report

Target: 74.207.244.221 | Tool: Nmap 7.99
Internship Task 2 | Analyst: Kali Linux

1. Scan Overview

Five Nmap scans were executed against the target host 74.207.244.221 (li86-221.members.linode.com) from a Kali Linux machine on 2026-05-30. Each scan used different flags to enumerate open ports, services, and OS information.

Target IP	74.207.244.221
Hostname	li86-221.members.linode.com
Nmap Version	7.99
Scan Date	2026-05-30 05:52 - 05:54 UTC-4
Host Status	Up (< 0.2s latency)
Network Distance	1 hop
Provider	Linode (Cloud Hosting)

2. Individual Scan Results

Scan 1: Service Version Detection on Port 110 (nmap -sV -O -p 110)

Command: nmap -sV -O -p 110 74.207.244.221

Duration: 3.74 seconds

Port 110 was found in a filtered state (no-response), preventing OS fingerprinting. Aggressive OS guesses included network cameras, print servers, wireless routers, and Cisco IOS devices — all at 89-92% confidence. No exact OS match was determined.

Port Scanned	110/tcp
Port State	filtered pop3
OS Guess (Top)	AXIS 2100 Network Camera / D-Link DP-300U (92%)
OS Guess (Other)	Linux 2.6.8 (90%), Cisco 1812/3640/3700 IOS 12.4 (89%)
Exact OS Match	None (test conditions non-ideal)

Scan 2: Fast Port Scan (nmap -F)

Command: nmap -F 74.207.244.221

Duration: 6.28 seconds | 100 most common ports scanned
93 filtered ports (no-response) were not shown. The following ports were discovered:

PORT	STATE	SERVICE	NOTES
22/tcp	open	ssh	SSH remote access
80/tcp	closed	http	HTTP web (closed)
110/tcp	open	pop3	POP3 email retrieval
143/tcp	open	imap	IMAP email access
465/tcp	open	smtps	SMTPS secure mail submission
993/tcp	open	imaps	IMAPS secure IMAP
995/tcp	open	pop3s	POP3S secure POP3

Scan 3: Aggressive Scan on Port 22 (nmap -A -p 22)

Command: nmap -A 74.207.244.221 -p 22
Duration: 22.47 seconds
Port 22 (SSH) was found in a filtered state. The aggressive scan also performed a traceroute, revealing a single-hop path to the target host.

Port Scanned	22/tcp
State	filtered ssh
Traceroute Hop 1	98.40 ms 221.244.207.74.in-addr.arpa (74.207.244.221)
Network Distance	1 hop

Scan 4: TCP SYN Stealth Scan (nmap -sT)

Command: nmap -sT 74.207.244.221
Duration: 100.62 seconds | All 1000 common ports scanned
All 1000 scanned TCP ports returned in ignored (filtered) states with no-response. No ports were reported as open during this scan. This may indicate aggressive firewall filtering on the target.

Ports Scanned	1000 (default common ports)
Result	All 1000 ports in ignored/filtered state
Note	Host confirmed up (0.092s latency)

Scan 5: Full Port Scan (nmap -p-)

Command: nmap -p- 74.207.244.221
Duration: 403.67 seconds | All 65,535 TCP ports scanned

This comprehensive scan enumerated all 65,535 ports. 64,895 filtered (no-response) and 634 filtered (net-unreachable) ports were found. The following open/closed ports were confirmed:

PORT	STATE	SERVICE	NOTES
22/tcp	open	ssh	SSH remote access
80/tcp	closed	http	HTTP web (closed)
110/tcp	open	pop3	POP3 email retrieval
143/tcp	open	imap	IMAP email access
465/tcp	open	smtps	SMTPS secure mail submission
993/tcp	open	imaps	IMAPS secure IMAP
995/tcp	open	pop3s	POP3S secure POP3

3. Consolidated Open Ports Summary

The table below consolidates all ports discovered across the fast scan (-F) and full scan (-p-):

PORT	STATE	SERVICE	DESCRIPTION
22/tcp	open	ssh	Secure Shell — remote login/command execution
80/tcp	closed	http	HTTP — port closed, no web server listening
110/tcp	open	pop3	Post Office Protocol v3 — email retrieval (unencrypted)
143/tcp	open	imap	Internet Message Access Protocol — email (unencrypted)
465/tcp	open	smtps	SMTP over TLS/SSL — secure mail submission
993/tcp	open	imaps	IMAP over SSL — encrypted email access
995/tcp	open	pop3s	POP3 over SSL — encrypted email retrieval

4. Analysis & Observations

4.1 Service Profile

The target host is primarily configured as a mail server. The combination of POP3 (110), IMAP (143), SMTPS (465), IMAPS (993), and POP3S (995) strongly indicates an email hosting service running on Linode cloud infrastructure. SSH (22) is open for remote administration.

4.2 Security Observations

- Unencrypted mail ports (110/POP3 and 143/IMAP) are open alongside their TLS counterparts — credentials transmitted over these ports are susceptible to interception.

- Port 80 (HTTP) is closed, suggesting no standard web server is running or the service has been deliberately disabled.
- Port 22 (SSH) appeared filtered in the -sV and -A scans but open in -F and -p- scans, which may indicate rate-limiting or scan-rate-based filtering.
- The -sT connect scan (Scan 4) showed all 1000 ports as filtered, suggesting a stateful firewall or intrusion prevention system may be active.
- OS fingerprinting was inconclusive across all scans — the host does not respond in a way that allows reliable OS detection.

4.3 OS Fingerprint

Nmap could not produce an exact OS match. Aggressive guesses ranged from network cameras and print servers to Cisco IOS routers and Linux 2.6.x kernels, all at 89-92% confidence. These low-confidence guesses are typical when firewalls drop or alter packets used in OS detection probes.

4.4 Network Topology

Traceroute from the -A scan showed a single hop to the target (98.40 ms RTT), confirming the host is directly reachable from the scanner without intermediate hops being visible — typical of Linode cloud VPS instances.

5. Scan Commands Reference

COMMAND	PURPOSE	DURATION
nmap -sV -O -p 110 74.207.244.221	Version + OS detect, port 110	3.74 sec
nmap -F 74.207.244.221	Fast scan (100 common ports)	6.28 sec
nmap -A 74.207.244.221 -p 22	Aggressive scan, port 22	22.47 sec
nmap -sT 74.207.244.221	TCP connect scan (1000 ports)	100.62 sec
nmap -p- 74.207.244.221	Full scan (all 65535 ports)	403.67 sec

6. Conclusion

The reconnaissance conducted against 74.207.244.221 (Linode-hosted) reveals a mail server exposing both encrypted and unencrypted email protocols alongside SSH administration access. The presence of unencrypted POP3 and IMAP ports represents a potential security concern. OS detection was inconclusive due to firewall packet filtering. The host demonstrated active filtering behaviour during the TCP connect scan, suggesting defensive measures are in place.